

Wumpus No Longer Alone

1. Background

The classic Wumpus World pits a lone explorer against a single lurking beast. That era is over.

You have stumbled into an ancient cave system — one that has grown restless. Multiple Wumpuses now stalk its corridors in synchronized, oscillating patterns. Pits yawn in the darkness. Strange temporal anomalies warp your sense of time. The gold still gleams somewhere ahead, but the question is no longer *"Can I survive?"* — it is *"Is survival even guaranteed before I take a single step?"*

2. Problem Statement

You are given an $n \times n$ cave grid. An agent begins at cell $(1, 1)$ and must reach the cell containing the **Gold (G)**. Before the agent moves, you must determine whether a **guaranteed-safe path** to the gold exists, accounting for Wumpus movement over time, pits, time zones, and stench accumulation.

If such a path exists, find:

- (a) The **optimal path** that minimizes total time cost.

3. Grid Specification

- The grid is **1-indexed**: cell $(1, 1)$ is the top-left; cell (n, n) is the bottom-right.
- $5 \leq n \leq 20$.
- The agent moves **orthogonally** (up, down, left, right), one cell per turn.
- $(1, 1)$ and the cell containing **G** are always guaranteed safe — no Wumpus, Pit, or Time Zone will ever occupy these cells.

Each cell contains exactly one of the following symbols:

Symbol Name		Description
.	Empty	Safe to enter; no effect.
W	Wumpus	Deadly . Entering a cell occupied by a Wumpus kills the agent immediately.
P	Pit	Entering incurs a +5 second time penalty.
T	Time Zone	Entering reduces the agent's accumulated time by 3 seconds (a benefit).
G	Gold	The goal cell. Reaching it ends the journey successfully.

4. Perception Rules

The agent perceives adjacent hazards **without entering the cell**. Percepts appear in the four orthogonally adjacent cells of a hazard.

4.1. Stench

A Wumpus at cell (i, j) emits **Stench** into all of its orthogonal neighbours:

$$(i - 1, j), \quad (i + 1, j), \quad (i, j - 1), \quad (i, j + 1)$$

The Wumpus's own cell does **not** carry a stench marker — only its neighbours do.

4.2. Breeze (Perceived)

A Pit at cell (i, j) emits a **perceived Breeze** into all of its orthogonal neighbours:

$$(i - 1, j), \quad (i + 1, j), \quad (i, j - 1), \quad (i, j + 1)$$

Percepts are **additive**: a cell adjacent to both a Wumpus and a Pit carries both Stench and perceived Breeze simultaneously. The agent may use these percepts for logical inference.

4.3. Stench Accumulation — Death Rule

If the agent passes through **3 or more stench-bearing cells** over the course of a path (cells that are adjacent to a Wumpus *at the time of entry*), **the agent dies**.

Important: Stench is **dynamic** and evaluated at the exact turn the agent enters a cell.

5. Wumpus Movement

Wumpuses move **horizontally only**, following a strict oscillating pattern determined by their **current row index** i at the start of each turn pair. Movement is bounded by the grid perimeter — if a Wumpus would move beyond the boundary, it **stays at the boundary cell** for the remainder of that step.

5.1. Turn Structure

Movement alternates between **Turn A** and **Turn B**, cycling indefinitely.

When i is even	
Phase	Steps
Turn A	Move right 3, then left 1
Turn B	Move right 1, then left 3
When i is odd	
Phase	Steps
Turn A	Move left 3, then right 1
Turn B	Move left 1, then right 3

Let the net displacement per full cycle (Turn A + Turn B) be:

$$\Delta_{\text{even}} = (+3 - 1) + (+1 - 3) = 0 \quad \Delta_{\text{odd}} = (-3 + 1) + (-1 + 3) = 0$$

The oscillation is **zero-net**: over any complete cycle the Wumpus returns to its original column (modulo boundary clamping).

Step-by-step resolution: Each individual step within a turn is resolved sequentially. If the Wumpus hits a wall mid-sequence, it halts at the boundary and resumes from that position for subsequent steps in the same turn.

Row tracking: The row i used to determine movement type is evaluated **at the beginning of each turn**. Wumpuses move strictly horizontally and never change rows.

Turn order: The Wumpus's position is updated **before** the agent moves each turn.

6. Time Cost Model

The agent's journey accumulates time $\mathcal{T}$ according to the following rules:

Event	Time Effect
Moving into any cell (base cost)	+1 second
Entering a Pit P	+5 seconds
Entering a Breeze cell B	+2 seconds
Entering a Time Zone T	-3 seconds
Entering a Wumpus cell W	Agent dies
Stench count ≥ 3	Agent dies

Formally, after each move the accumulated time is updated as:

$$\mathcal{T} \leftarrow \max(0, \mathcal{T} + \delta_{\text{cell}})$$

where δ_{cell} is the net cost of the entered cell. Time is clamped to zero and **cannot go negative**.

7. Constraints

Parameter	Bound
Grid size n	$5 \leq n \leq 15$
Number of Wumpuses	$0 \leq W \leq n$
Number of Pits	$0 \leq P \leq \lfloor n^2/5 \rfloor$
Number of Time Zones	$0 \leq T \leq \lfloor n^2/5 \rfloor$
Turn limit	$4 \times n^2$

A path exceeding $4n^2$ turns is considered **invalid** (the agent is presumed lost).

8. Input Format

```
n
grid row 1
grid row 2
...
grid row n
```

- The grid is given as n space-separated rows, each containing n symbols.
- Exactly one cell contains G.
- Cell (1,1) is always . (the agent's starting position).
- There may be zero or more Wumpuses.

Example Input

```
5
. . . P .
W . . . .
. P . . G
. . W . .
. . . T .
```

9. Output Format

```
Line 1:  SAFE  or  UNSAFE
```

If **UNSAFE**, print only UNSAFE and terminate.

If **SAFE**, print:

```
Line 1:  SAFE
Line 2:  <minimum time cost>
Line 3:  <optimal path as sequence of (row,col) pairs>
```

The path on Line 3 must be printed as space-separated coordinate pairs:

```
(1,1) (1,2) (1,3) (1,4) (2,4) (3,4) (3,5)
```

If multiple paths share the minimum cost, print the **lexicographically smallest** one (compare coordinate pairs left to right).

Example Output

```
SAFE
3
(1,1) (1,2) (1,3) (1,4) (1,5) (2,5) (3,5)
```

10. Notes & Clarifications

1. **Guaranteed-safe path:** There exists a sequence of moves such that, given the deterministic Wumpus movement, the agent reaches **G** without dying.
2. A path through a **Pit** is valid — it incurs a penalty but does *not* kill the agent.
3. The **stench death rule** counts unique stench-bearing cells entered, not the number of Wumpuses nearby. A cell adjacent to two Wumpuses simultaneously counts as *one* stench encounter.
4. Two paths are **distinct** if they differ in at least one cell visited or in the order of visiting.
5. The agent **cannot revisit** a cell within a single path (no cycles).
6. The Gold cell **G** is always safe and incurs no cost beyond the base movement cost of +1.

11. Evaluation Criteria

1. **Proof & Algorithm Choice (20%)**
Justification for chosen algorithm; time and space complexity analysis in context of the state space.
2. **Correctness (30%)**
Optimal path found; all constraints enforced — Wumpus movement, stench death, time clamping, and boundary behaviour.
3. **Visualization (20%)**
Interactive simulation showing agent movement, dynamic stench fields, and Wumpus positions per turn
4. **Code Quality (15%)**
Modular and readable; logic and display cleanly separated; variables and data structures self-explanatory at a glance.

5. Test Cases & Innovation (15%)

Coverage of edge cases; bonus credit for analytics dashboards, custom heuristics, or creative visualization of dynamic elements.

6. The teams must submit:

- a. GitHub repository
 - b. Proof documentation (*e.g. LaTeX document*)
 - c. Video explanation (*max 3 minutes*)
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